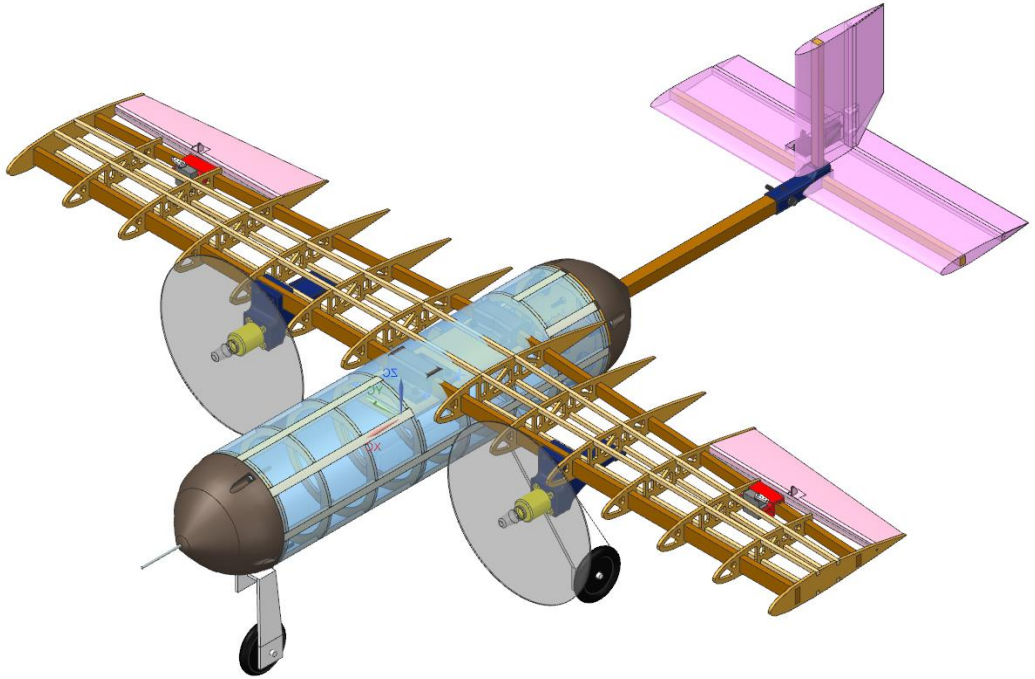
Senior Capstone: Aircraft Design



Problem: Design, build, and test RC aircraft capable of carrying payload and recording critical flight data

Method:

- Design from mission requirements, MATLAB based math models, and Siemens NX modeling
- Manufacture with laser cut plywood, custom 3D printed mounts and brackets
- Transfer aerodynamic loads across airframe via secure attachments, bolts, fasteners

Solution:

- Functional aircraft capable of stable flight without feedback control (full manual piloting)
- Successful data collection allowing for analysis of anomaly during flight and comparison to predicted performance

CFD Analysis: Supersonic Inlet

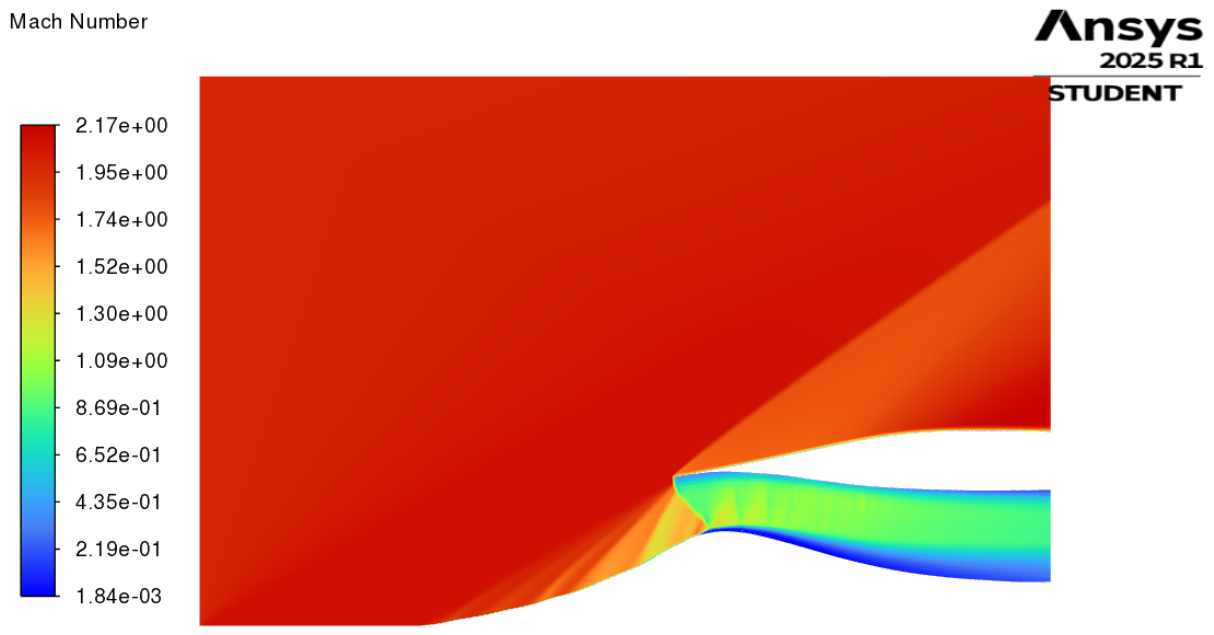
Goal: Perform CFD analysis on supersonic inlet under different flow cases

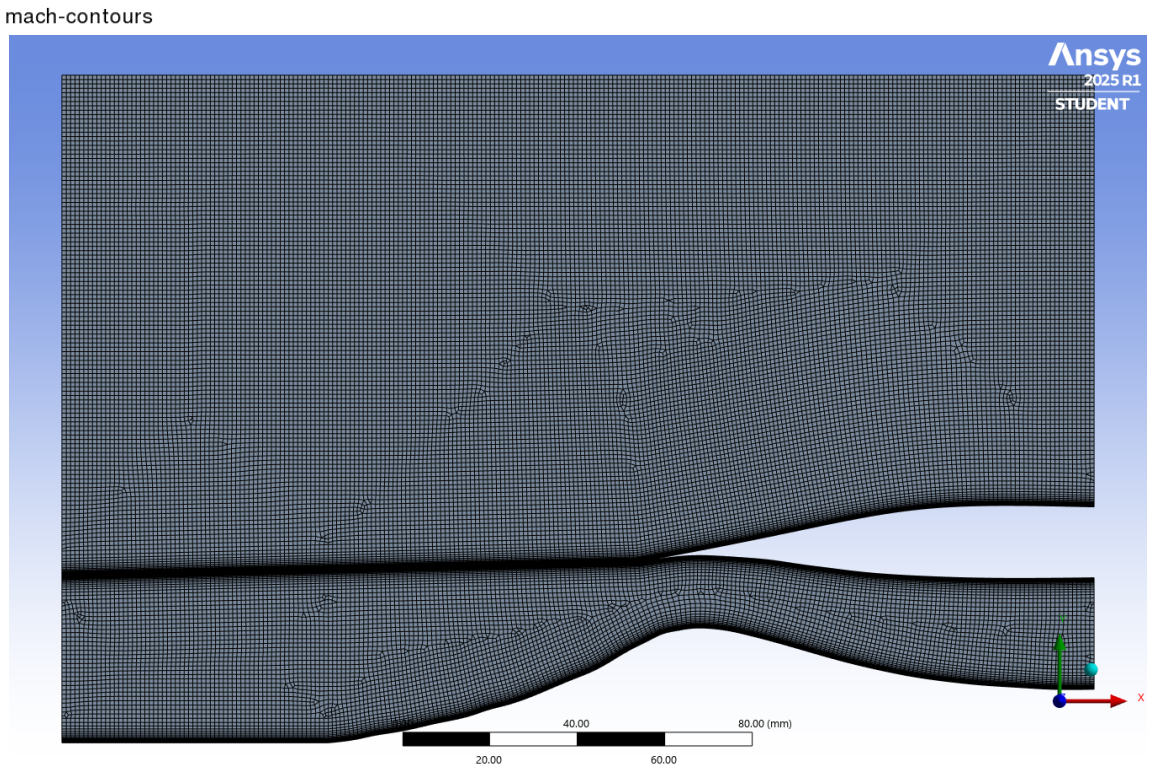
Method:

- Define reference flow and obtain solutions on coarse grid with appropriate y+ value
- Perform grid independence study by creating multiple grids
- Analyze using density-based solver with second order methods and AGR to resolve shock locations

Result:

- Estimated inlet performance in off-design conditions (non-zero angle of attack leeward, low Mach number cases)
- Grid refinement study demonstrating tradeoff between solution accuracy and computational cost





Firefighting RC Aircraft

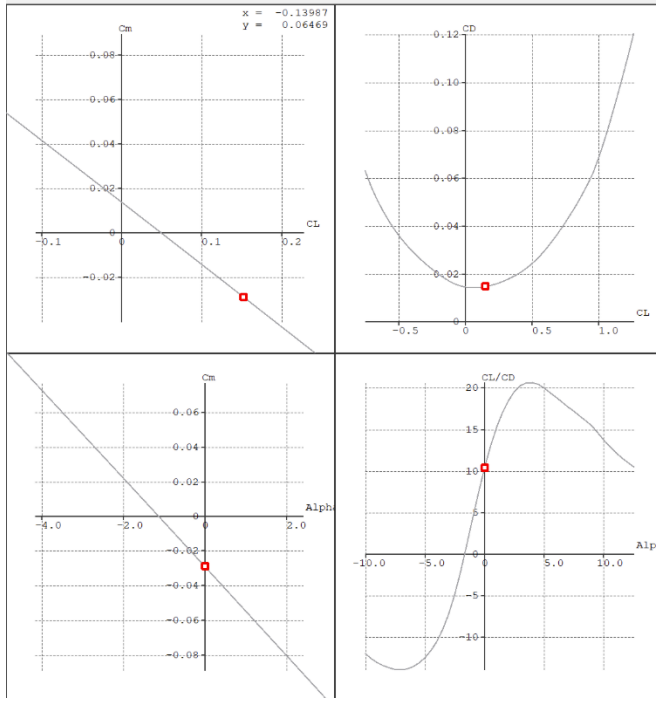
Problem: Create a prototype that supports the integration of electronics that collects data on surroundings

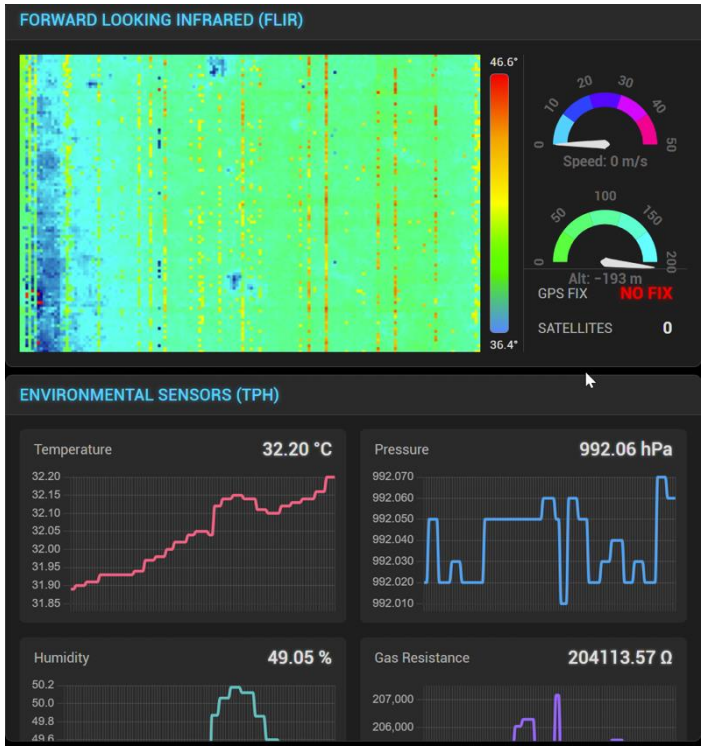
Method:

- Design from first principles and mission requirements, revise based on initial flight test data and stability simulations
- Build using lightweight and durable material
- Solder connectors and wires to support electronics

Solution:

- Working prototype capable of smooth flying
- Successful integration of TPH sensor and thermal camera for live telemetry and data collection





FEA: Integrated Truss Structure

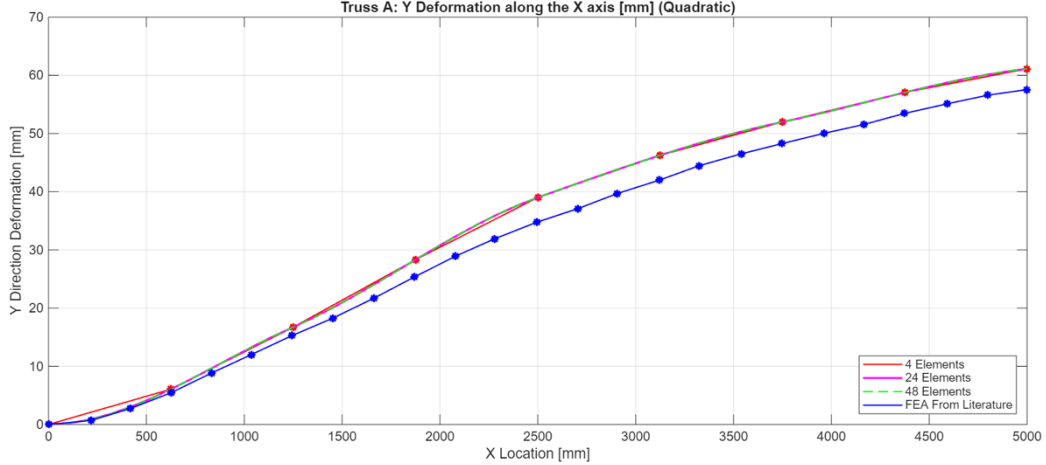
Goal: Predict and validate the structural deformation of an integrated truss under varying load scenarios using both analytical and computational methods

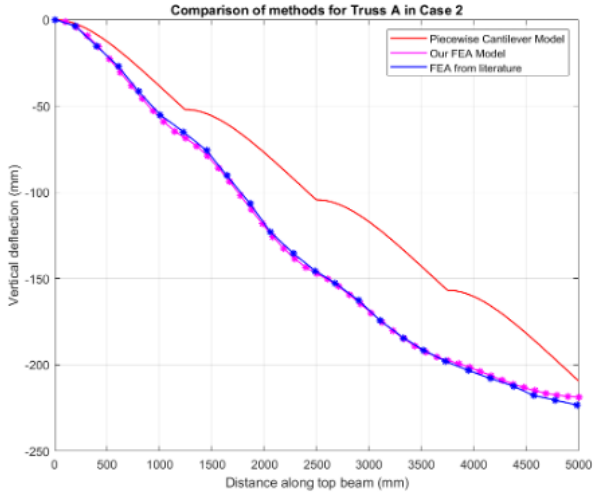
Method:

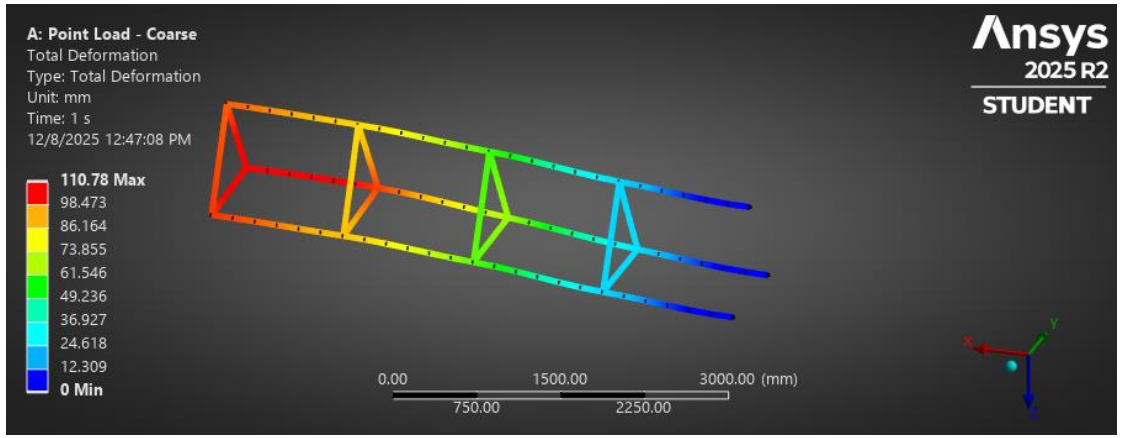
- Derive first-order approximation from Euler-Bernoulli beam theory
- Develop finite-element model using ANSYS FEA suite
- Obtain solutions for each load scenario

Result:

- Comparison of simple approximation with full degree of freedom finite-element model and solution from literature
- Validated accurate deformation prediction across all load scenarios, confirming model fidelity







Conceptual Aircraft Design

Problem: Produce a conceptual design to meet customer requirements

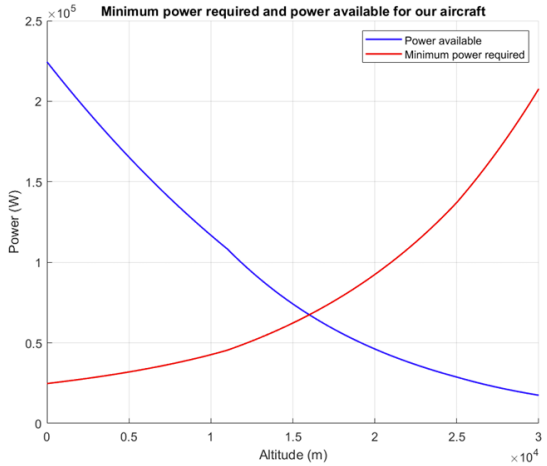
Method:

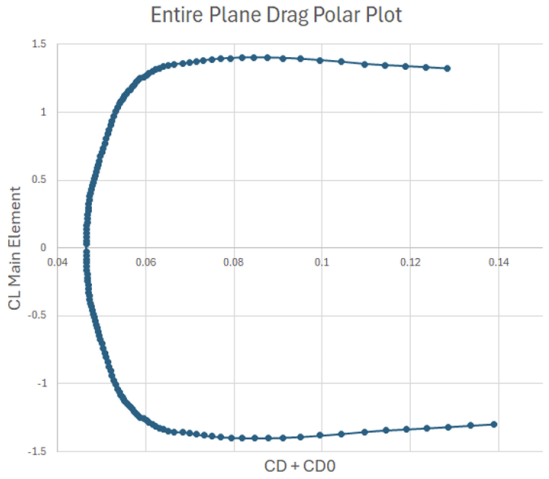
- Define mission profile and requirements
- Choose configuration based on historical data and weighted decision matrices
- Iteratively size the aircraft using MATLAB scripts

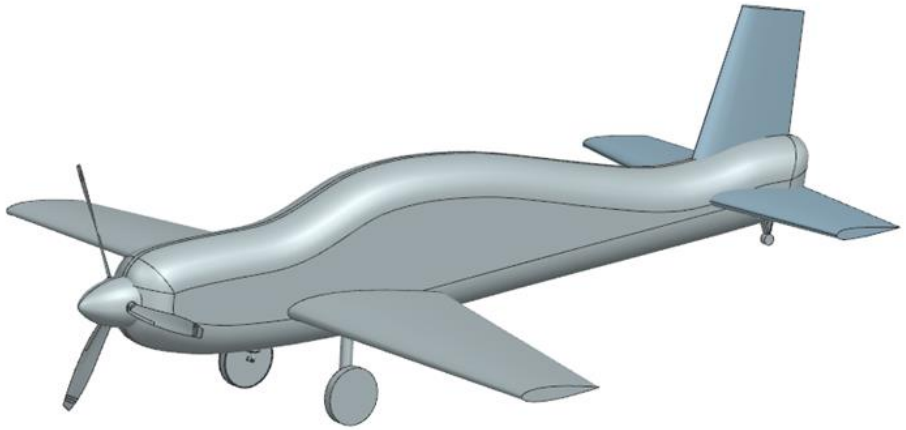
Solution:

- Converged, conceptual aircraft configuration optimized for the mission
- Comprehensive technical report detailing overall design process and outlining next steps in prototyping and testing

Parameter	Symbol	Value
Wingspan	b	6.12 m
Wing area	S	6.192 m^2
Wing aspect ratio	AR_w	6.05
Tip chord	c_{tip}	0.675 m
Root chord	c_{root}	1.35 m
MAC	c_{bar}	1.05 m
Winglets?	N/A	No
Landing gear type?	N/A	Fixed
Prop. Efficiency	η_p	0.86
Empty weight	m_{empty}	517.5 kg
MTOW	m_{TO}	683 kg
Length	l	5.17 m
Height	h	1.97 m
V-Stab area	S_{VT}	0.816 m^2
H-Stab area	S_{HT}	1.5 m^2





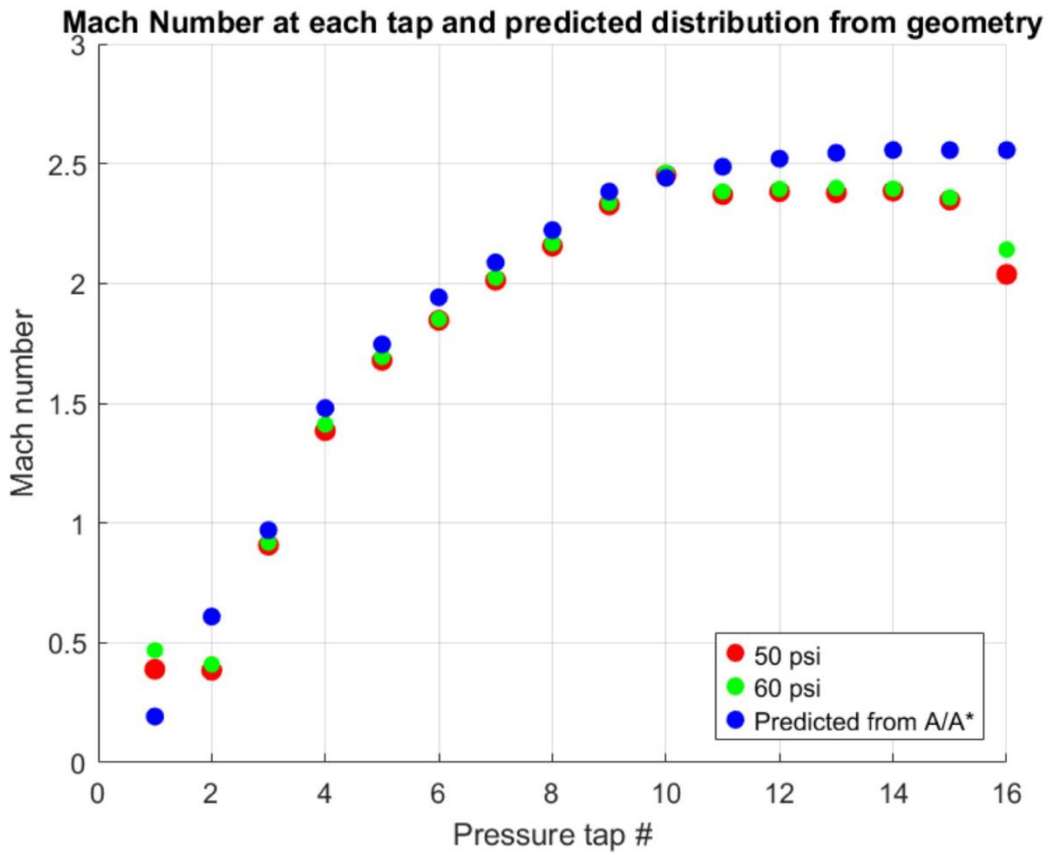


Supersonic Wind Tunnel Testing

Problem: Investigate shock formation over bodies in supersonic flow

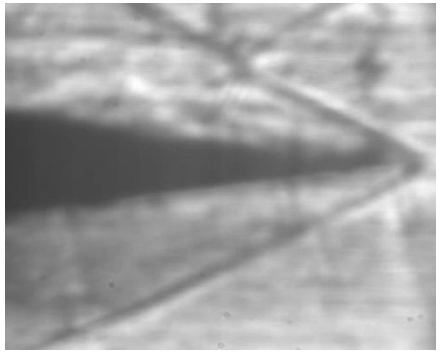
Method:

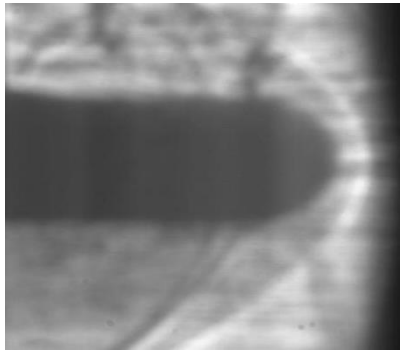
- Calibrate supersonic tunnel and verify freestream Mach number
- Take schlieren images of each geometry to visualize shock formation
- Collect flow field data using pressure taps



Results:

- Verified wind tunnel performance
- Validated exact oblique shock relations by comparing experimental results to theory
- Obtained clear schlieren imagery over bodies commonly used in aerospace vehicle design



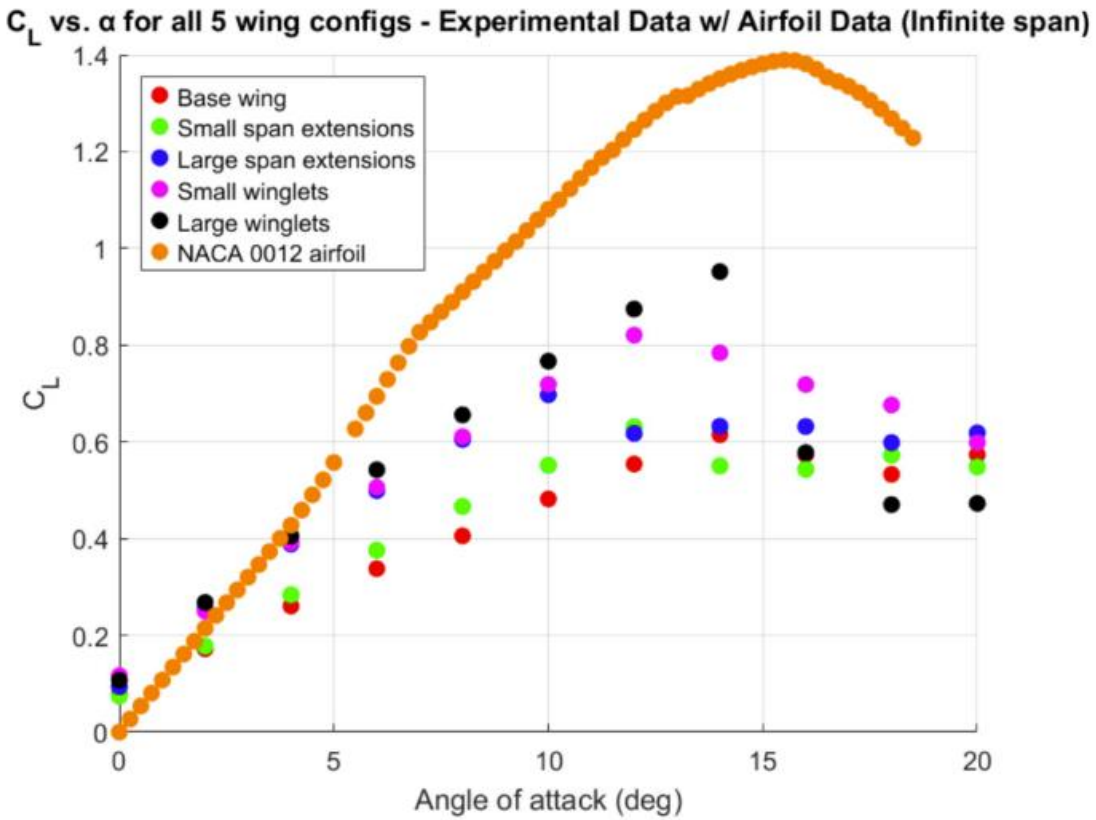


Finite Wing Wind Tunnel Test

Problem: Compare aerodynamic data of wing to that of its airfoil

Method:

- Select an airfoil and create model of wing
- Configure and operate low-speed wind tunnel for testing
- Collect data of various wing spans and winglet configurations using load cells



Results:

- Visualized the effect of wing-tip vortices on lift
- Demonstrated how winglets can improve aerodynamic performance
- Increased understanding of how error can impact experimental data

Rocket Fin Analysis

Problem: Investigate how fin geometry affects aerodynamic drag on a model rocket during subsonic flight conditions.

Method:

- Design and vary fin geometries using Siemens NX, controlling sweep angle, aspect ratio, and planform shape
- Operate a low-speed wind tunnel to simulate launch-phase airflow and collect drag measurements via load cell instrumentation

Results:

- Captured drag coefficient data across all fin configurations, revealing clear aerodynamic performance trends
- Synthesized findings into a technical report including comparative analysis and design recommendations

Figure 1.1: Model rocket without fins

Figure 1.2: Model rocket with small fins

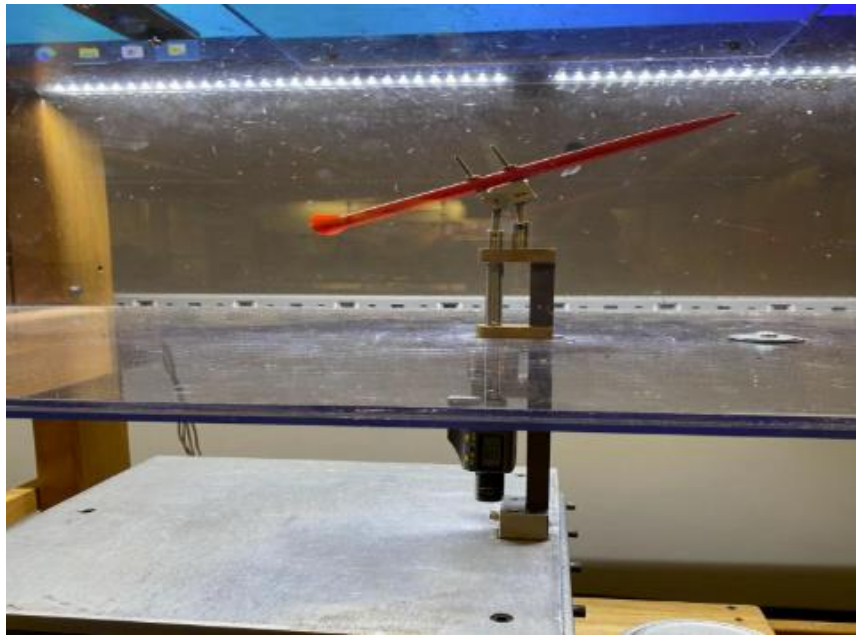



Figure 1.3: Model rocket with large fins